Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/27/2020
PI Name:	Lawler, John Ph.D.		
Project Title:	Redox Regulation of nNOS Translocation and Muscle Atrophy During Mechanical Unloading		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/24/2012	**End Date:**	11/15/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	4
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 11/15/2018 (latest end date was 8/22/2018 before this extension) per F. Hernandez/ARC (Ed., 10/16/18) NOTE: Extended to 8/22/2018 (previously extended in 2016 to 8/22/2017) perNSSC information (Ed., 10/6/17) NOTE: Extended to 8/22/2017 per F. Hernandez/ARC and NSSC information (Ed., 8/30/16) NOTE: Extended to 8/23/2016 per A. Chu/ARC and NSSC information (Ed., 6/16/15) NOTE: Grant number changed to NNX13AE45G, per A. Chu via NSSC (Ed., 1/29/13) NOTE: Grant number changed to NNX12AR62G, per A. Chu via NSSC (Ed., 9/28/12) NOTE: Period of performance (previously 6/27/12-6/26/15) and grant number (previously NNX12AL23G) changed per A. Chu/ARC and NSSC information (Ed., 9/12/2012)		
Key Personnel Changes/Previous PI:	2018 report: Jeff Hord received his PhD. He is currently an NIH Research Fellow with Kevin Campbell's laboratory at the University of Iowa.		
COI Name (Institution):	Fluckey, James Ph.D. (Texas A&M University)		
Grant/Contract No.:	NNX13AE45G ; NNX12AR62G (previous grant #)		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	A critical limitation to astronaut performance during spaceflight missions including those to the International Space Station (ISS) and Mars is the impaired contractile ability of skeletal muscles. Skeletal muscle is a highly specialized tissue that rapidly adapts to dynamic changes in mechanical loading by modifying its ultrastructure and mass. Detection and transmittance of loading are vital to muscle integrity and protein turnover, with a nexus associated with the subsarcolemmal cytoskeleton that includes the dystrophin-glycoprotein complex (DGC) and caveolae. This interface between myocytes and the extracellular matrix confers mechanical integrity and activates cell signaling pathways that regulate protein turnover, mitochondrial, muscle fiber cross-sectional area, and fiber phenotype. Mechanical unloading experienced in the microgravity of spaceflight elicits large decrements in force generating capacity via reduction in muscle fiber cross sectional area or atrophy. While unloading-induced atrophy is a function of reduced protein synthesis coupled with increased protein degradation, activation of the transcription factors FoxO3a and nuclear factor-kappaB have drawn significant attention as mediators. Although the upstream triggers are not fully understood, translocation of neuronal nitric oxide synthase (nNOS) from the DGC and sarcolemma to the sarcoplasm is a novel signaling event that stimulates proteolysis, and thus atrophy. New clues can be gleaned from Duchenne and many of the limb-girdle muscular dystrophies, pathologies involving mutations to DGC, and other sarcolemmal proteins. Because MDs and microgravity both increase susceptibility of muscles to damage, oxidative stress, and pro-inflammatory signaling, a shared root cause is suspected: disruption of the subsarcolemmal cytoskeleton. New studies and Preliminary Data indicate the importance of NAD(P)H oxidase (NOX), a sarcolemmal source of oxidative stress, and caveolin-3 in stimulating myopathy in muscular dystrophy models. NOX may be activated upstream by angiotensin II and c-abl, a tyrosine kinase. Further, our pilot data show that unloading-induced alterations in caveolin-3 and nNOS are redox dependent and linked to atrophy and shift in muscle fiber-type. Our central hypothesis is that oxidative stress produced by NOX pathways and mitochondria directly contribute to a sarcolemmal loss of nNOS during mechanical unloading, enhancing proteolysis, and suppressing protein synthesis. The rodent hindlimb unloading model and novel fractional synthesis and degradation will be used to test our hypotheses. Specific Aim 1 will identify NOX-dependent signaling on nNOS translocation and muscle fiber atrophy with hindlimb unloading. Specific Aim 2 will determine directly the role of mitochondrial and cytosolic oxidative stress in movement of nNOS away from the muscle cell membrane and disruption of DGC with mechanical unloading. In short, our primary objective is to identify the trigger by which skeletal muscles switch from being an anabolic to a catabolic machine during spaceflight, causing muscle atrophy and weakness. We propose that oxidative stress produced by a NOX pathway and mitochondria move the master switch protein (nNOS) away from the cell membrane, which slows protein synthesis and accelerates breakdown of protein. Our study would directly address high-priority gaps in "Recapturing a Future for Space Exploration" including the role of reactive oxygen species in protein balance and maintaining the slow contractile phenotype. Our grant proposal would also forward our understanding of (1) alterations in the subsarcolemmal cytoskeleton and membrane domains, (2) sensing and transduction of loading and gravity, (3) protein turnover, and (4) regulatory mechanisms that govern alterations in skeletal muscle, all programmatic emphases and goals of the current Space Biology NRA (NASA Research Announcement).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bedrest, casting, and spaceflight cause significant skeletal muscle atrophy and weakness as a result of disuse. Prolonged spaceflight presents great challenges to the physical performance and health of astronauts as a result of microgravity, or reduced gravitational forces (i.e., loads). Mechanical unloading and extreme disuse during spaceflight produce rapid and dramatic skeletal muscle and weakness. Prolonged disuse also predisposes skeletal muscle to damage when landing on Mars, in an EVA (extra-vehicular activity), or returning to Earth. Furthermore, chronic disuse also contributes to and exacerbates underlying chronic pathology, including insulin resistance and cardiovascular disease. As a result crew health, safety, and completion of mission objectives are placed in jeopardy. While the pathways that cause atrophy and weakness due to disuse remain uncertain, proteins that sense changes in forces that skeletal muscles must produce may be the key. One of these proteins is nitric oxide synthase (nNOS), which is positioned near the muscle cell membrane. During spaceflight as well as aging and some types of muscular dystrophy, nNOS appears to be transported away from the cell membrane where it transforms from a muscle building to a muscle wasting agent. We will focus on oxidative stress, produced by excess free radicals produced by proteins near the cell membrane (ex. NOX) and mitochondria and how they affect nNOS transport as well as skeletal muscle atrophy. Understanding how these key membrane proteins are regulated will not only aid us in our treatment of disuse-related atrophy, but also will allow development of new therapeutics combating sedentary lifestyle and possible application to aging, Type 2 diabetes, and muscular dystrophies. Indeed, changes that are altered during spaceflight and ground models of spaceflight also appear to change with aging and diseases that cause skeletal muscle wasting and weakness. For example, loss of nNOSμ from the sarcolemma is a commonality abound spaceflight/disuse, aging, Type 2 diabetes, and Duchenne muscular dystrophy. Oxidative stress appears to be at the root of nNOSμ loss in theses muscle myopathies. Out laboratory seeks to understand the underlying redox mechanisms that leader to perturbation in nNOS function and thus myopathy.
	Skeletal muscle is a highly specialized tissue that adapts rapidly to changes in mechanical loading by altering its cross-sectional area and mass. The reduced mechanical loading of spaceflight causes postural muscles in the lower extremities and flexors in the upper extremities to lose fiber cross-sectional area and force-generating capacity. Muscle atrophy and weakness may then impede mission objectives, increase the risk of injury, and increase susceptibility to damage upon reloading. Muscular alterations with spaceflight are therefore a critical limitation to performance during spaceflight missions and a serious health concern for astronauts. Excessive reactive oxygen species (ROS) production is considered an important component of muscle myopathy with unloading. ROS are produced from the mitochondria and non-phagocytic NADPH-oxidases (NOX) in skeletal muscle fibers and thought to contribute to atrophy through activation of proteolytic pathways. In addition, ROS may initiate a signaling pathway that results in disruption of the dystrophin-glycoprotein-complex (DGC). The DGC serves as an interface between myocytes and the extracellular matrix and is responsible for the detection and transmittance of load necessary for regulation of various cellular processes. Recent evidence from our laboratory and others indicates that ROS-induced atrophy is related to disruption of the DGC and translocation of the mu(μ)-splice variant of nNOS from

Task Progress:	the sarcolemma into the cytosol. nNOSμ is then thought to directly stimulate FoxO3a and ubiquitin ligases (MuRF-1, atrogin-1), leading to muscle fiber atrophy. The purpose of this investigation is to identify oxidant-related mechanisms that lead to disruption of the DGC, translocation nNOSμ, proteolysis, and thus atrophy. Our central hypothesis is that ROS, produced by NOX pathways and mitochondria, cause translocation of nNOSμ from the sarcolemma during mechanical unloading, enhancing proteolysis and suppressing protein synthesis. Our experimental design consists of a series of studies utilizing a rodent hindlimb unloading model of spaceflight to: (1) identify NOX dependent signaling of nNOS translocation, FoxO3a activation, and muscle fiber atrophy, and (2) determine the redox regulatory role in DGC disruption with mechanical unloading and atrophy. In addition, additional investigation has been commenced to better translate these findings to spaceflight, where we have begun to use pharmaceutical (Losartan) and nutraceutical (fish oil + curcumin) interventions to limit unloading-induced atrophy. Data that has been collected thus far indicates that: (1) administration of a potent antioxidant, the SOD/catalase mimetic EUK-134, prevents unloading induced atrophy during a short-term (54 hr) unloading protocol. Reduced atrophy was related to decreased expression of NOX subunits, ROS production, and markers of ROS-related signaling/damage. In addition, nNOS translocation was prevented with EUK-134 administration as well as related activation of proteolysis through FoxO3a dephosphorylation; (2) EUK-134 also prevented unloading induced atrophy during a longer 7-day unloading protocol, an effect also related to decreased ROS production and nNOS translocation as well as activation of anabolic pathways; (3) More specific targeting of ROS production by using a NOX-2 inhibitor, gp91 ds-tat, and mitochondrial ROS inhibitor, SS-31, has also resulted in reduced muscle fiber atrophy following a 7-day unloading protocol. Interestingly, an interaction between these two sources of ROS has also been suggested with our pilot data; and lastly, (4) pharmaceutical inhibition of angiotensin II receptor I (AT1R), and (5) a nutraceutical interventions with fish oil + curcumin, respectively, have demonstrated similar results. In summary, inhibition of ROS appears to be effective in limiting skeletal muscle atrophy with mechanical unloading. Inhibition of ROS is related to augmented anabolic and reduced catabolic signaling as well as translocation of nNOS, and the source of ROS production may be from both the mitochondria and Nox2 isoform of NADPH oxidase. Future studies will focus on finalizing the 7-day EUK-134 interventional study including measurement of fractional synthesis rate; further delineating the mechanisms responsible for reduced atrophy with peptidyl inhibition of ROS using gp91ds-tat to target Nox2 and SS-31 to target mitochondria. We will also utilize a genetic ablation of Nox2 (gp91phox-/-) to further establish its role in unloading induced atrophy; and establishing the effectiveness of pharmaceutical and nutraceutical interventions in preventing unloading-induced atrophy. In addition, fractional synthesis rate (FSR) and rate of protein degradation will be assessed in future experiments.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Articles in Peer-reviewed Journals	Lawler JM, Garcia-Villatoro EL, Guzzoni V, Hord JM, Botchlett R, Holly D, Lawler MS, Janini Gomes M, Ryan P, Rodriguez D, Kuczmarski JM, Fluckey JD, Talcott S. "Effect of combined fish oil & curcumin on murine skeletal muscle morphology and stress response proteins during mechanical unloading." Nutrition Research. 2019 May;65:17-28. Epub 2019 Jan 7. https://doi.org/10.1016/j.nutres.2018.12.013 ; PMID: 30954343 , May-2019
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